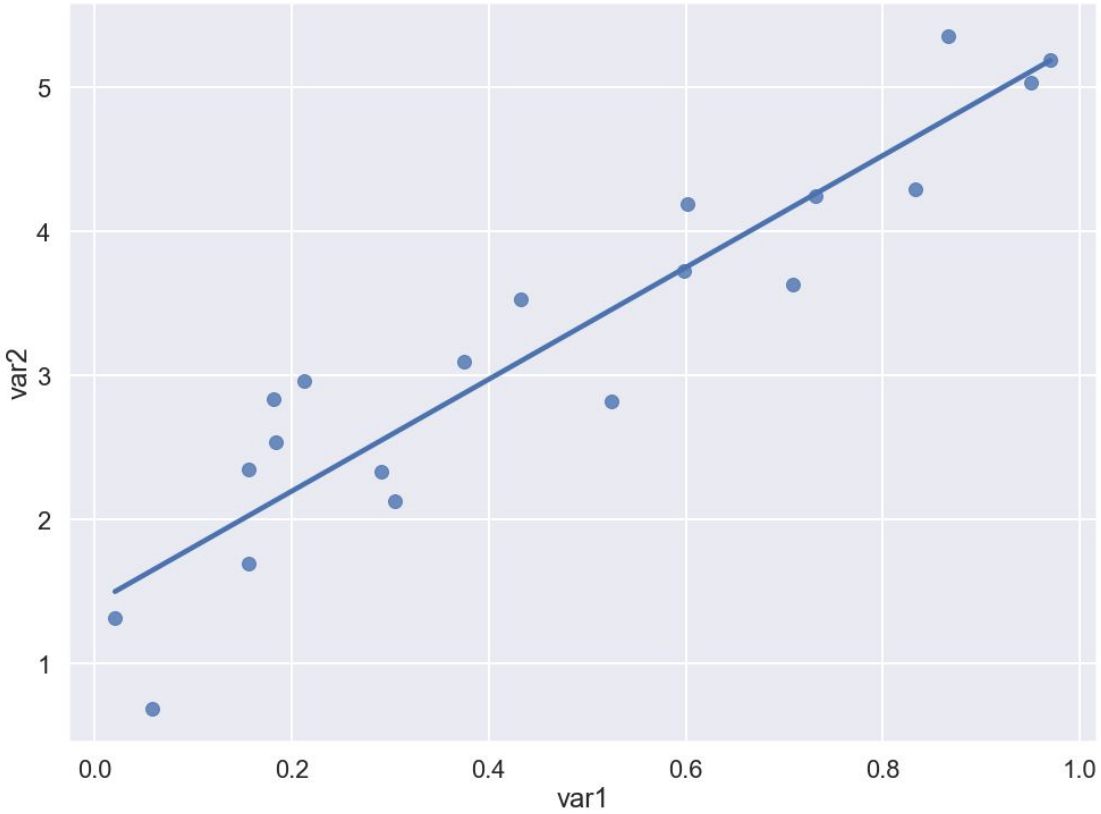


How to (not) give a presentation

WiSe26/27

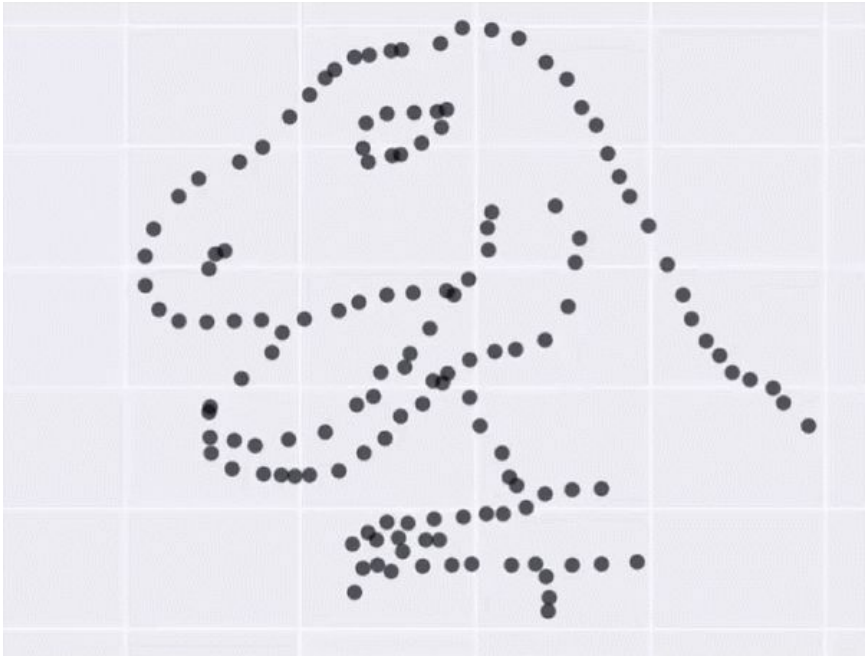
Graphs - Why visualizations?

Helps to understand data/research



var1	var2
0.375	3.096
0.951	5.033
0.732	4.244
0.599	3.726
0.156	1.692
0.156	2.35
0.058	0.69
0.866	5.359
0.601	4.19
0.708	3.633
0.021	1.318
0.97	5.191
0.832	4.292
0.212	2.959
0.182	2.84
0.183	2.534
0.304	2.13
0.525	2.819
0.432	3.528
0.291	2.336

What is common among these diverse looking datasets?



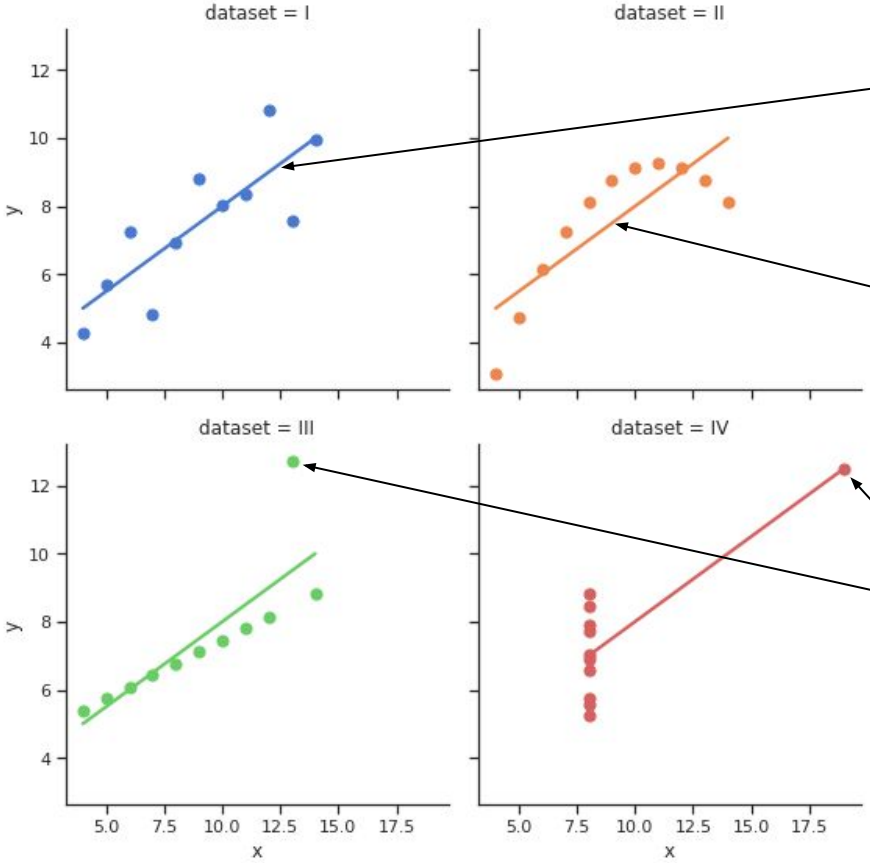
Answer: Summary statistics for all the datasets are exactly the same!

X Mean	:	54.26
Y Mean	:	47.83
X SD	:	16.76
Y SD	:	26.93
Corr.	:	-0.06

Never trust summary statistics alone

Dataset: *Cairo, A. Download the Datasaurus: Never trust summary statistics alone; always visualize your data. http://www.thefunctionalart.com/2016/08/download-datasaurus-never-trust-summary.html*

Helps to identify problems



Suited to fit a regression line

Not suited to fit a regression line

Outliers?

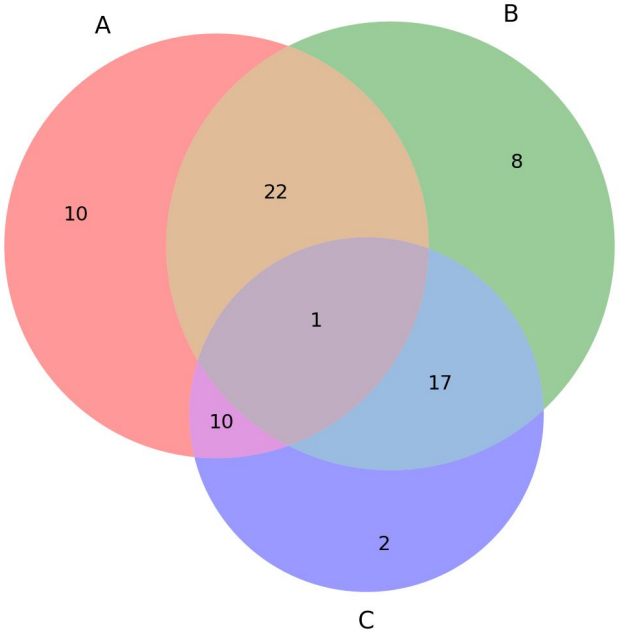
Dataset: *Anscombe, F. J. (1973). Graphs in statistical analysis. The american statistician, 27(1), 17-21.*

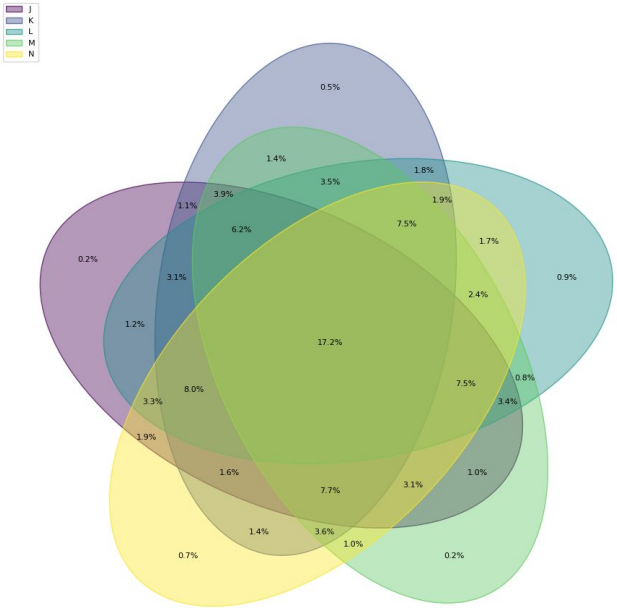
What not to do?

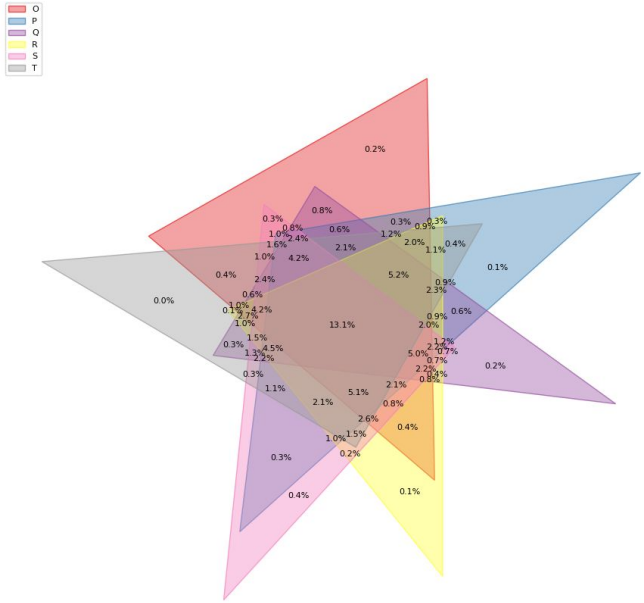
Venn Diagrams

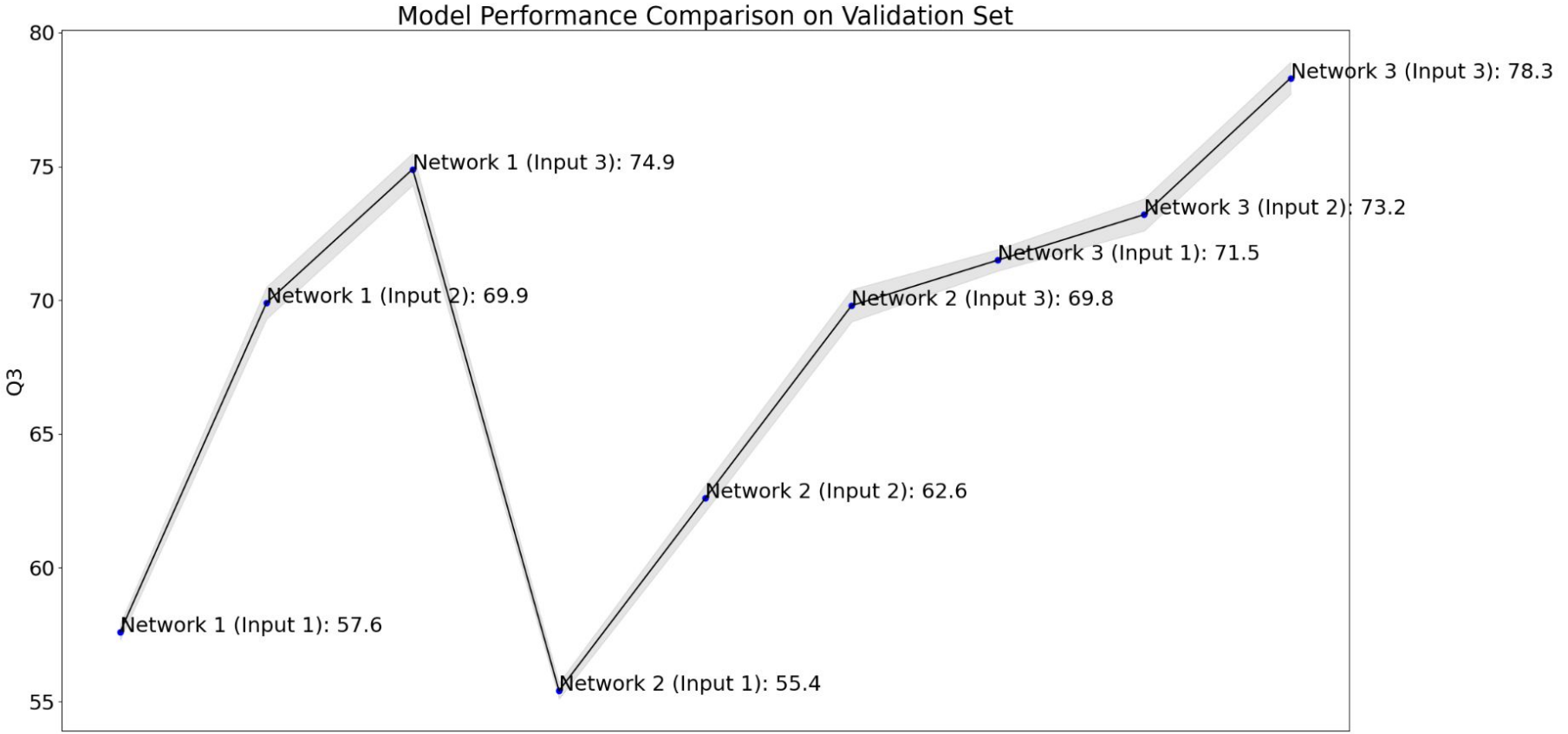
Misleading: section size ≠ sample size, hard to find intersection of interest

Alternatives: (stacked) bar plot, upset plot, heatmap








Model Performance Comparison on Validation Set
Q3
80
75
70
65
60
55
Network 1 (Input 1): 57.6
Network 1 (Input 2): 69.9
Network 1 (Input 3): 74.9
Network 2 (Input 1): 55.4
Network 2 (Input 2): 62.6
Network 2 (Input 3): 69.8
Network 3 (Input 1): 71.5
Network 3 (Input 2): 73.2
Network 3 (Input 3): 78.3

Text and highlighting

Proteins are **large** biomolecules or macromolecules that are comprised of one or more long chains of amino acid residues. Proteins perform a vast array of functions within organisms, including catalysing metabolic reactions, DNA replication, responding to stimuli, providing structure to cells and organisms, and transporting molecules from one location to another. Proteins differ from one another primarily in their sequence of amino acids, which is dictated by the nucleotide sequence of their genes, and which usually results in protein folding into a specific 3D structure that determines its activity.

A linear chain of amino acid residues is called a **polypeptide**. A protein contains at least one long **polypeptide**. Short **polypeptides**, containing less than 20–30 residues, are rarely considered to be proteins and are commonly called **peptides**, or sometimes **oligopeptides**. The individual amino acid residues are bonded together by **peptide** bonds and adjacent amino acid residues. The sequence of amino acid residues in a protein is defined by the sequence of a gene, which is encoded in the genetic code. In general, the genetic code specifies 20 standard amino acids; but in certain organisms the genetic code can include selenocysteine and—in certain archaea—pyrrolysine. Shortly after or even during synthesis, the residues in a protein are often chemically modified by post-translational modification, which alters the physical and chemical properties, folding, stability, activity, and *ultimately*, the function of the proteins. Some proteins have **non-peptide** groups attached, which can be called prosthetic groups or cofactors. Proteins can also work together to achieve a particular function, and they often associate to form stable protein complexes.

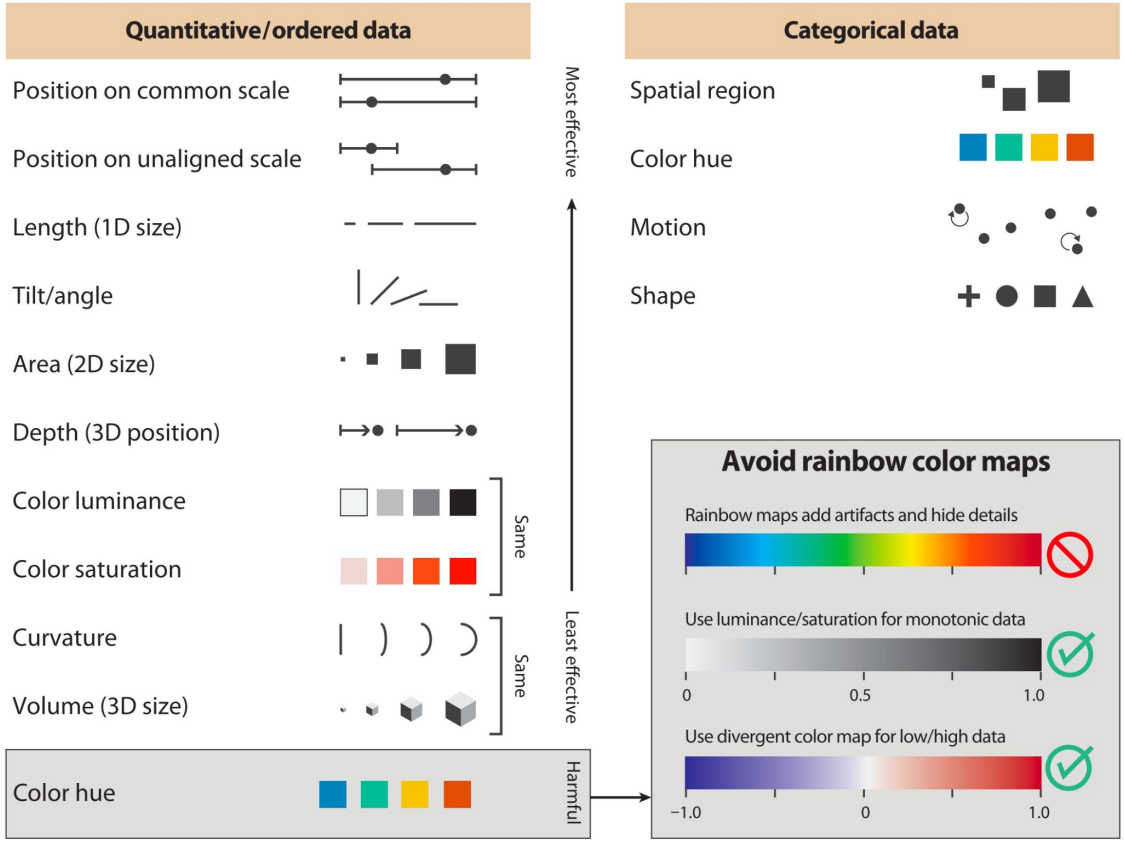
Once formed, proteins only exist for a certain period and are then degraded and recycled by the **cell's machinery** through the process of protein turnover. A protein's lifespan is measured in terms of its *half-life* and covers a wide range. They can exist for minutes or years with an average lifespan of 1–2 days in *mammalian* cells. Abnormal or misfolded proteins are degraded more rapidly either due to being **targeted** for destruction or due to being **unstable**.

Like other biological macromolecules such as **polysaccharides** and nucleic acids, proteins are essential parts of organisms and participate in virtually every process within cells. Many proteins are enzymes that catalyse biochemical reactions and are vital to **metabolism**. Proteins also have structural or mechanical functions, such as *actin* and *myosin* in muscle and the proteins in the ***cytoskeleton***, which form a system of **scaffolding** that maintains cell shape. Other proteins are important in cell signaling, immune responses, cell adhesion, and the cell cycle. In animals, proteins are needed in the diet to provide the essential amino acids that cannot be synthesized. Digestion breaks the proteins down for use in the **metabolism**.

Source: https://en.wikipedia.org/wiki/Protein

Data visualization concepts

Ranking of visual elements and common mistakes



Additional visual channels (unranked): connection, containment/enclosure, crispness/resolution, flicker, line endings, line pattern, line weight, numerosity, text, texture, transparency, weight/boldness.

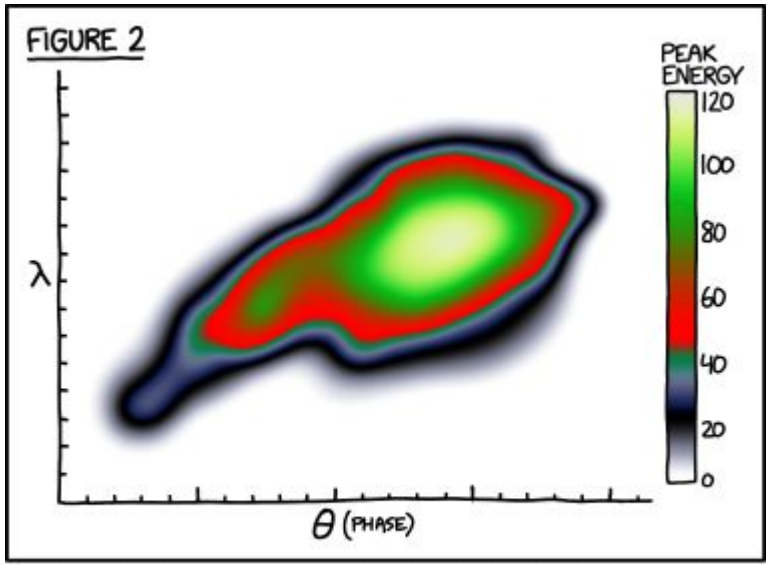
Things to avoid:

- rainbow colour maps
- Too much colour
- 3D visualization
- overcrowded visualizations
- conflate research and art

O'Donoghue, S. I., Baldi, B. F., Clark, S. J., Darling, A. E., Hogan, J. M., Kaur, S., ... & Procter, J. B. (2018). Visualization of biomedical data. Annual Review of Biomedical Data Science, 1, 275-304.

Rules of Thumb - Colors

- Colorblind friendly
- Easily distinguishable
- Document the meaning of color
- Consistent color choice
- Don’t add meaningless color
- Can you make it work in grayscale/ single color?



https://imgs.xkcd.com/comics/painbow_award.png

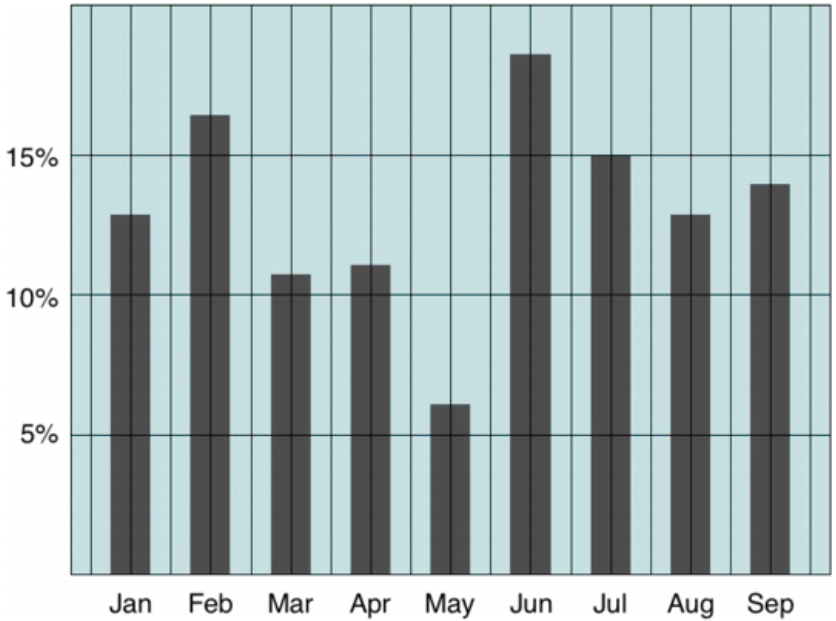
Data-to-Ink

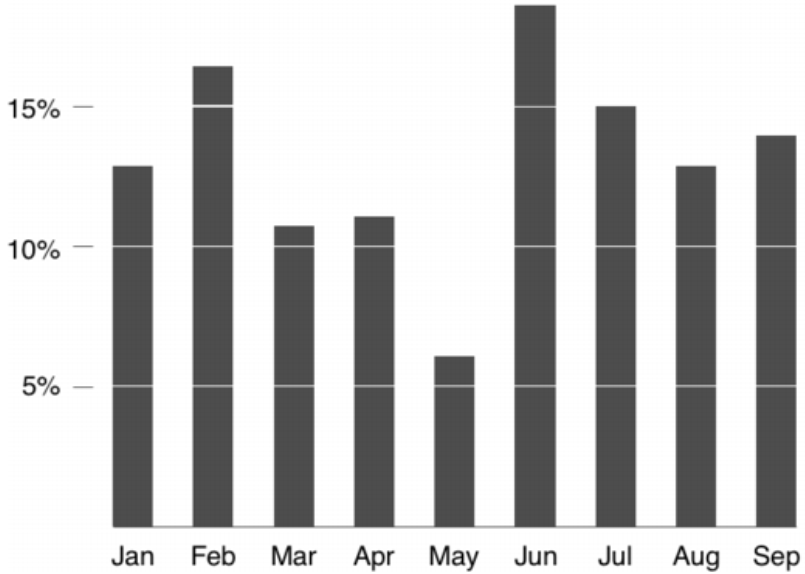
> Above all else, show the data. – *Edward R.Tufte*

$$\text{Data_to_Ink} = \frac{\text{Data_Ink}}{\text{Total_Ink}}$$

Edward R. Tufte, The Visual Display of Quantitative Information, 2nd ed., 8th print (Cheshire, Conn: Graphics Press, 2013).

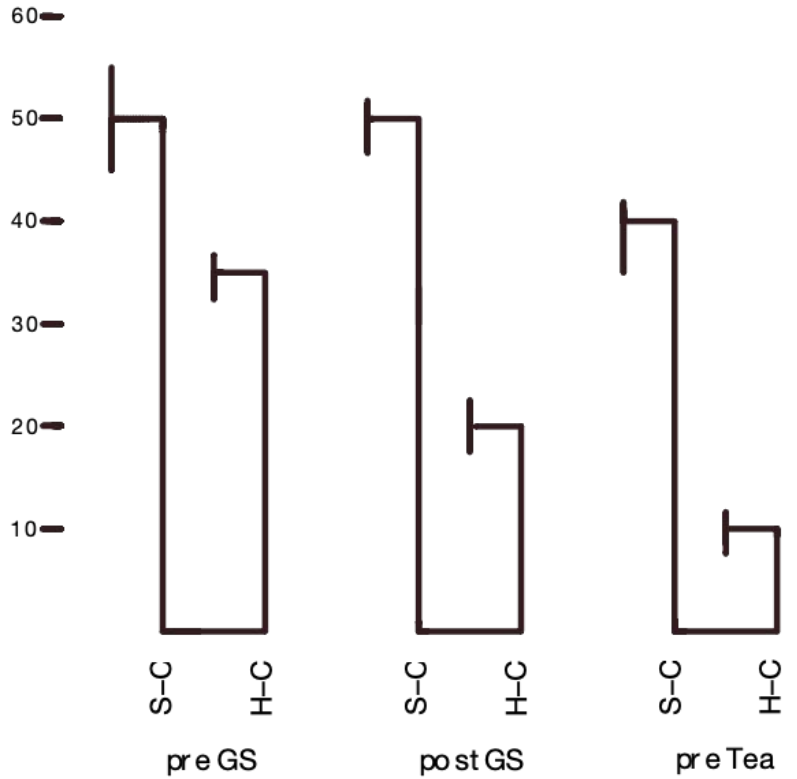
Data-to-Ink





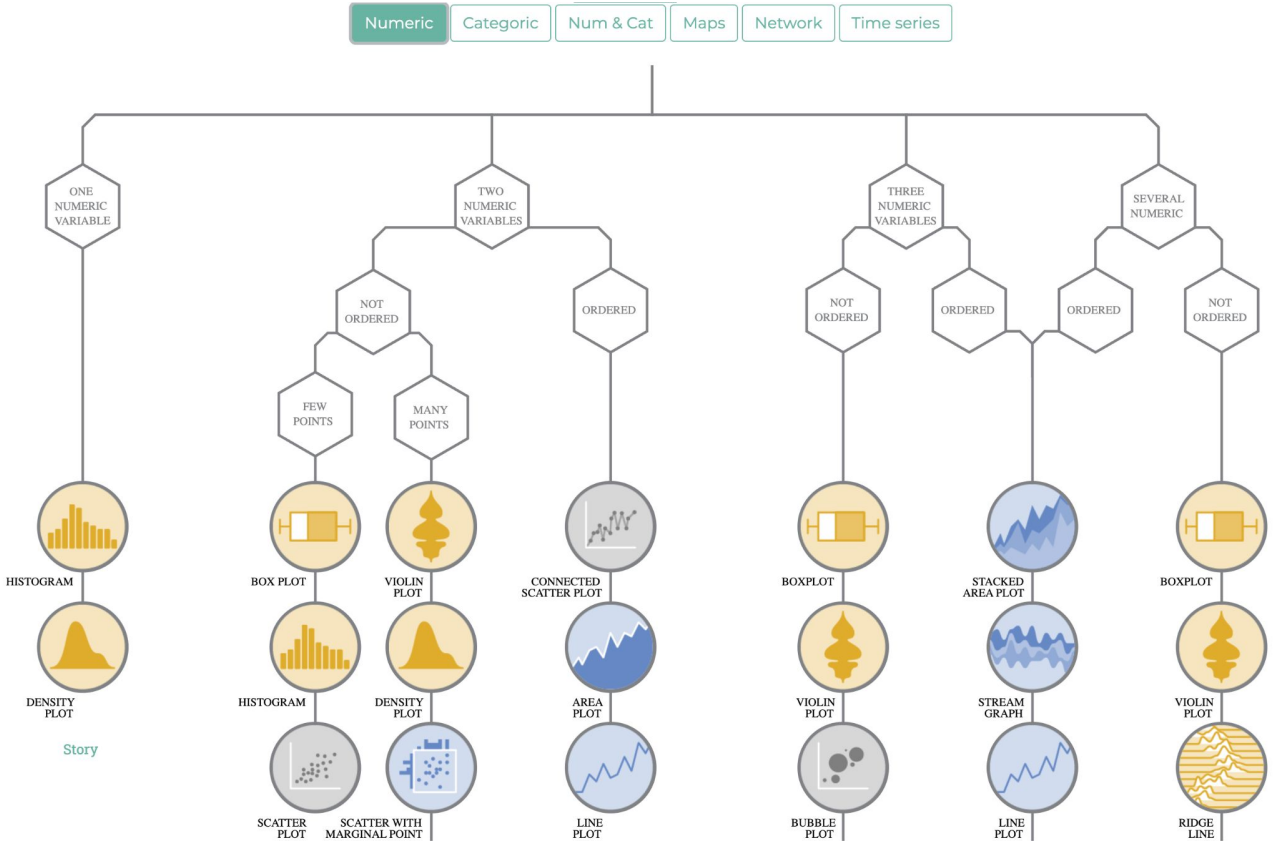
Edward R. Tufte, The Visual Display of Quantitative Information, 2nd ed., 8th print (Cheshire, Conn: Graphics Press, 2013).

How Little is too Little?



Plot choice

- Inspirations
- Plots for specific data types
- Caveats
- Example code in R, Python, D3.js

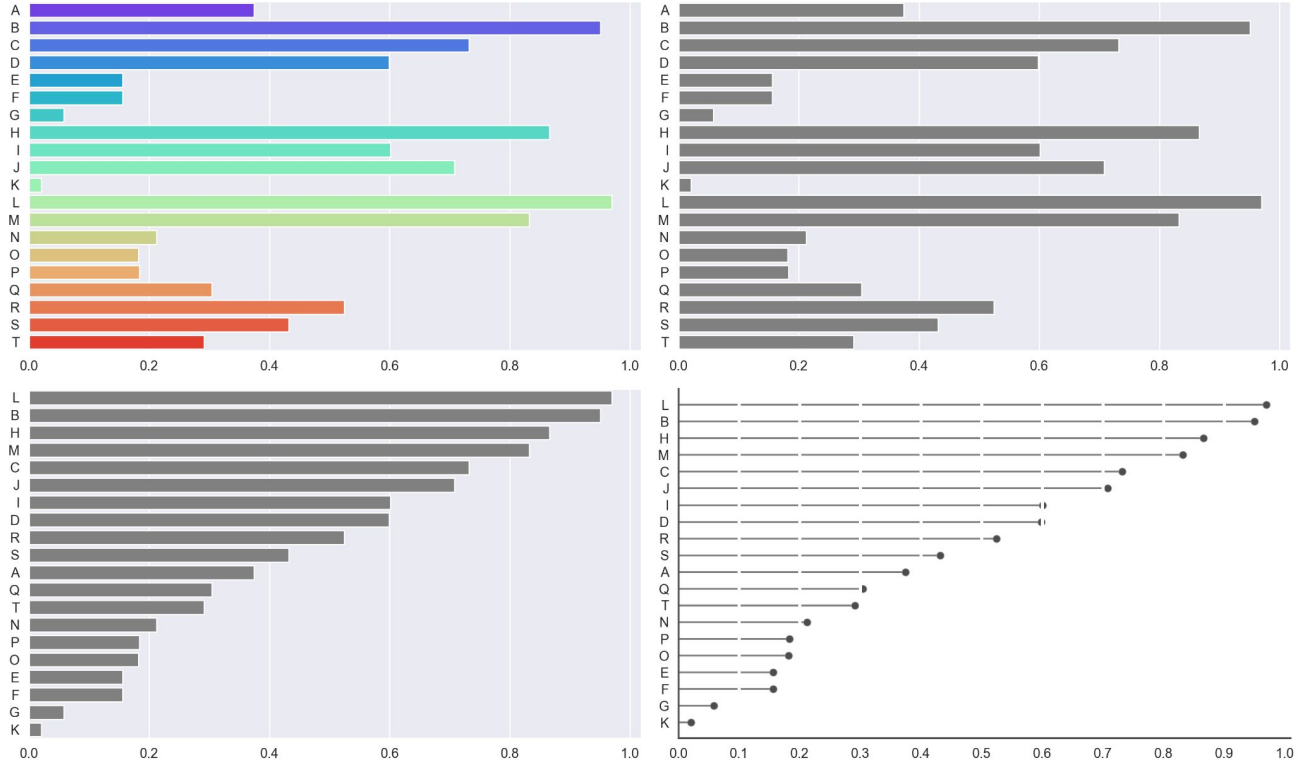


https://www.data-to-viz.com

Conveying Your Main Message

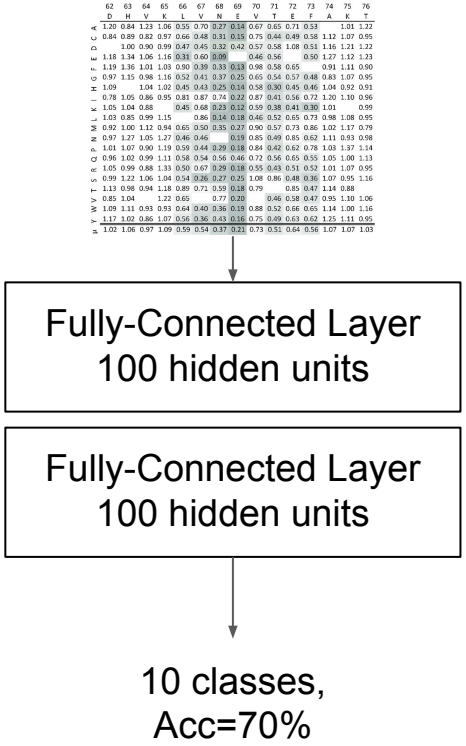
Bar plot & lollipop/stem plot

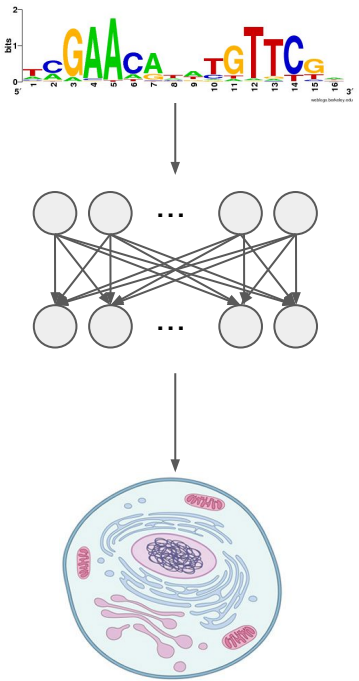
- Bars: no color + order by size
- Lollipop alternative to many bars



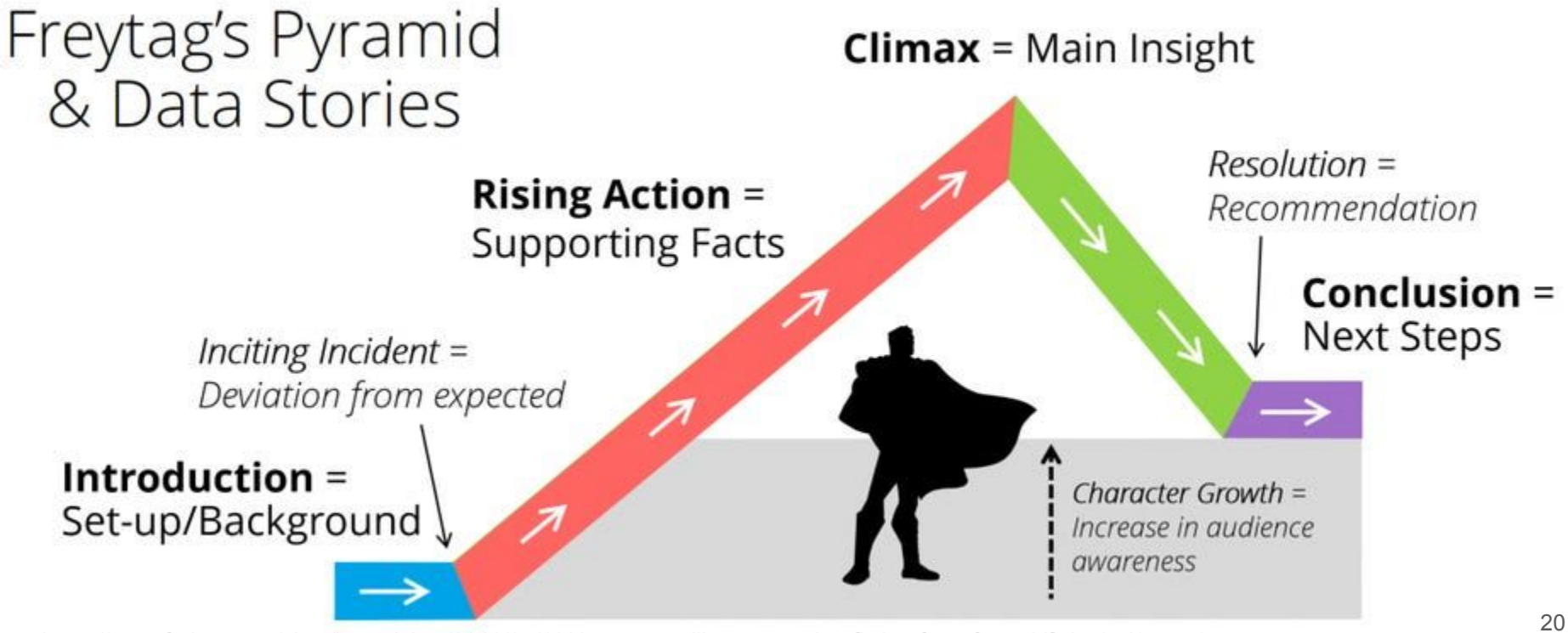
Visualization of workflows

- Input: PSSMs
- Trained ANN with 2 layers and 100 hidden units each
- Output: Predicted subcellular localization (10 classes)
- Achieved accuracy: 70%





Structuring your Presentation



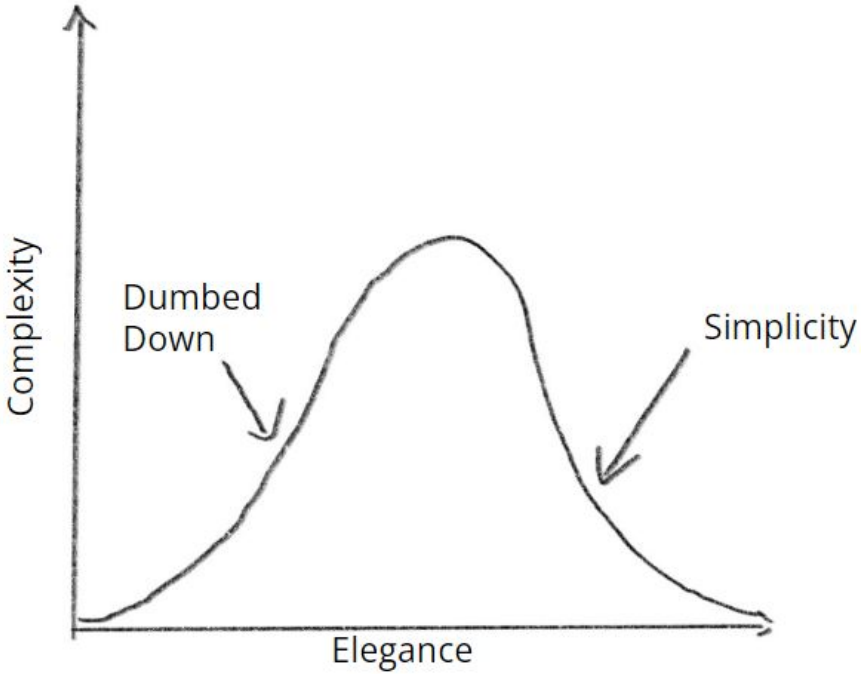
https://www.forbes.com/sites/brentdykes/2016/07/13/data-storytelling-separating-fiction-from-facts-2/?sh=1ad9ac4e3a7c

Rules of Thumb - Conveying your Main Message

- What does your audience know?
- What additional information is needed to understand your talk?
- What aspects of your work are most relevant?
- Are your claims/ conclusions supported by your presented data?
- Highlight important points on your visualizations (arrow/ use a pointer,...)

Keep it Simple!

- Focus on your main message
- Visualize as much as possible
- Few bullet points (max: 6), few words



Summary

- Mind colours, font size, visual elements
- Think about the message you want to convey
- Be consistent
- Keep it simple, less is more
- Give credits (aka. citing)
- Slide numbers